

Homework 04: Chapter 05

EEC 3150

For each Python exercise, write a separate Python script. Submit all scripts on Blackboard under the appropriate assignment. Name your script files with the following format:

HW<assignment number>_Ex<exercise number>_<FirstInitial><LastName>.py

Example: HW01_Ex01_GWest.py

For short answer or math exercises, either do your work on a sheet of paper and upload a picture of it to Blackboard or submit some kind of text file or Word document to Blackboard. You can place all such exercises in a single document as long as you clearly label each exercise. These files should follow a similar naming convention as the Python scripts.

Exercise 1 (10 points, Python)

1. Write a program which will read the file `CSVData.csv` and store its contents as a 2D list of floats. Each line should be a list in the larger 2D list. For example, if the file looked like this

```
1, 2, 3
4, 5, 6
7, 8, 9
```

then your 2D list should look like this

```
[[1, 2, 3],
 [4, 5, 6],
 [7, 8, 9]]
```

2. Print the 2D list with the following code:

```
for x in data:
    print(x)
```

(assuming `data` is the name of your 2D list).

Exercise 2 (20 points, Python)

1. Write a program (not a function) that will read the files

- `Mercury.txt`
- `Venus.txt`
- `Earth.txt`
- `Mars.txt`
- `Jupiter.txt`

and store the radius of each planet in a float variable. NOTE: in the first four files, the radius is called `Mean radius (km)` while in the fifth file it is called `Equat. radius (1 bar)`.

2. Write the radii to the file `PlanetRadii.txt` in the following format:

```
Mercury radius: 2440km
```

and so on with each planet having its own line.

Exercise 3 (20 points, Python)

1. Write a program (not a function) which will read the file `"bible-psalms.txt"` and create a nested dictionary `psalms`. The keys of this dictionary will be the chapter numbers and the values will be another dictionary. The keys of these inner dictionaries will be a verse number and the values will be a string that is the verse's text. NOTE: the tab (`\t`) escape character is used in each line of the file to separate the verse number and the verse text.
2. Give a demo of indexing a particular verse from the dictionary. Here is an example (you may choose your preferred chapter/verse):

```
chapter = 1
verse = 1
print(psalms[chapter][verse])
```

The output for the above example is:

```
Blessed is the man that walketh
not in the counsel of the
ungodly, nor standeth in the
way of sinners, nor sitteth
in the seat of the scornful.
```